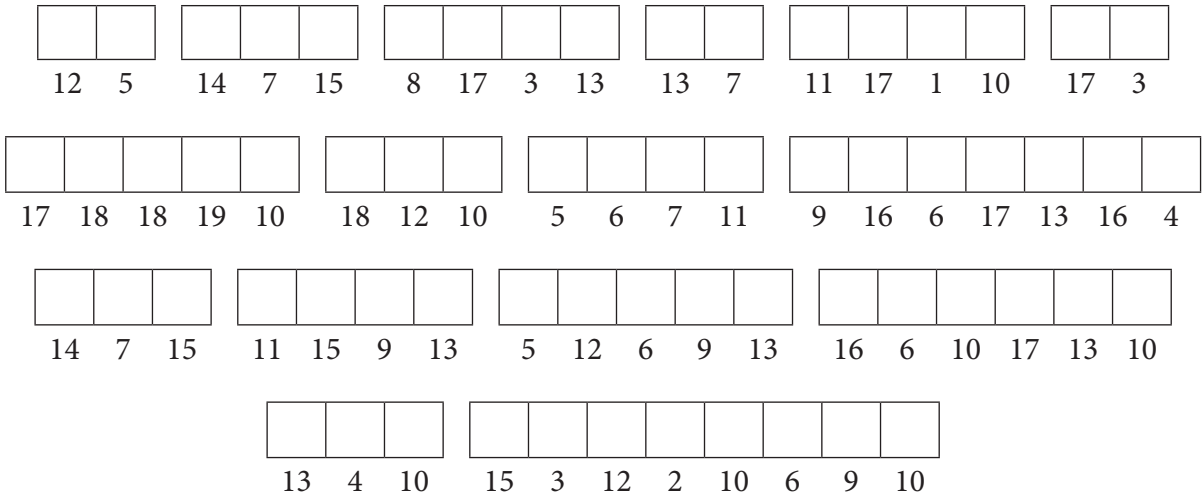
Name: _____

WORKSHEET

Rational and irrational numbers



Carl Sagan

Complete the questions on the following page, using the letters from the solutions to complete the well-known quote by Carl Sagan.

© Cengage Learning Australia Pty Ltd 2013

Questions

Classify the following numbers as rational or irrational.

1 $\frac{3}{11}$

2 $\sqrt{49}$

3 $\sqrt[3]{8}$

4 $\sqrt{2}$

5 -0.14

6 π

7 $\sqrt{\frac{100}{144}}$

8 $-70.\overline{4704}$

9 e

10 $\sqrt[3]{\frac{27}{40}}$

Are the following rational numbers recurring or terminating when expressed as decimals?

11 $\frac{8}{33}$

12 $\frac{324}{625}$

Convert the following recurring decimals into rational numbers, $\frac{a}{b}$, in simplest form.

13 $0.\dot{2}$

14 $0.\dot{5}$

15 $3.4\dot{7}$

16 $0.\overline{96}$

17 $2.3\overline{15}$

18 $0.\overline{841}$

19 $6.5\overline{773}$

Solutions

Solutions		
Question	**Rational**	**Irrational**
1	**K**	**D**
2	**V**	**J**
3	**N**	**A**
4	**P**	**H**
5	**F**	**X**
6	**U**	**R**
7	**O**	**B**
8	**W**	**Q**
9	**Y**	**S**
10	**G**	**E**

A $\frac{382}{165}$

C $\frac{32}{33}$

I Terminating decimal

L $\frac{32854}{4995}$

M Recurring decimal

P $\frac{841}{999}$

T $\frac{2}{9}$

U $\frac{313}{90}$

Y $\frac{5}{9}$

© Cengage Learning Australia Pty Ltd 2013